Attention and Transformers

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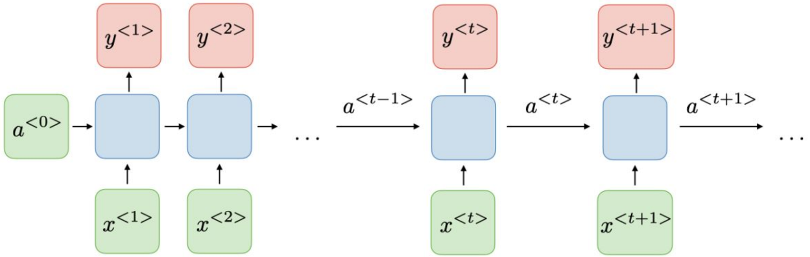
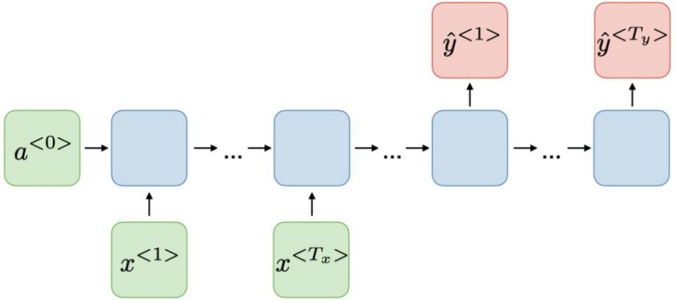
Background

Recurrent Neural Networks



Many-to-many
$T_x \neq T_y$

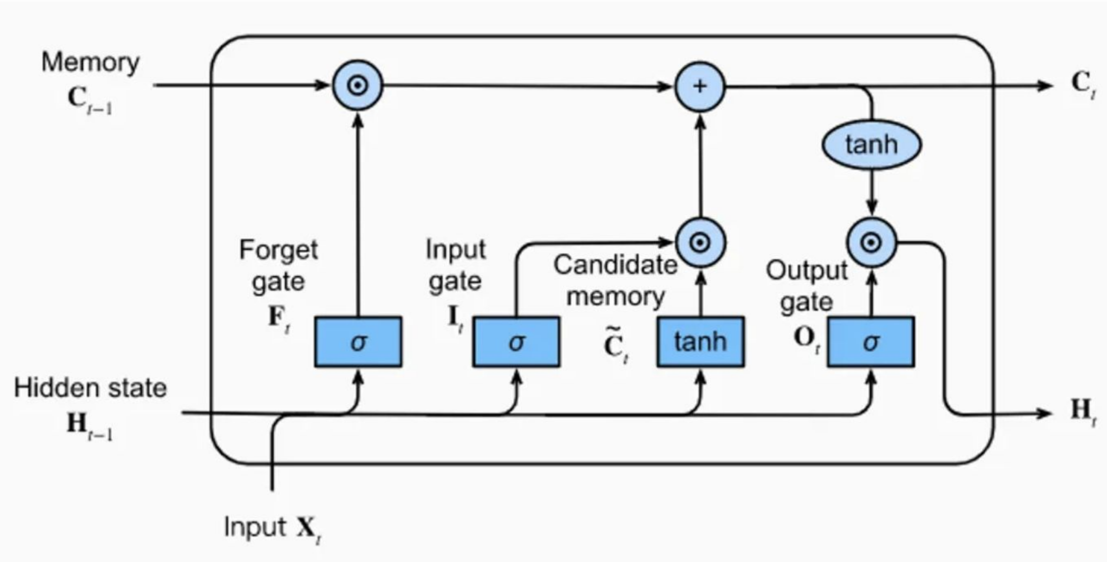


Machine translation

Vanilla RNN

$$\boxed{a^{<t>} = g_1(W_{aa}a^{<t-1>} + W_{ax}x^{<t>} + b_a)} \quad \text{and} \quad \boxed{y^{<t>} = g_2(W_{ya}a^{<t>} + b_y)}$$

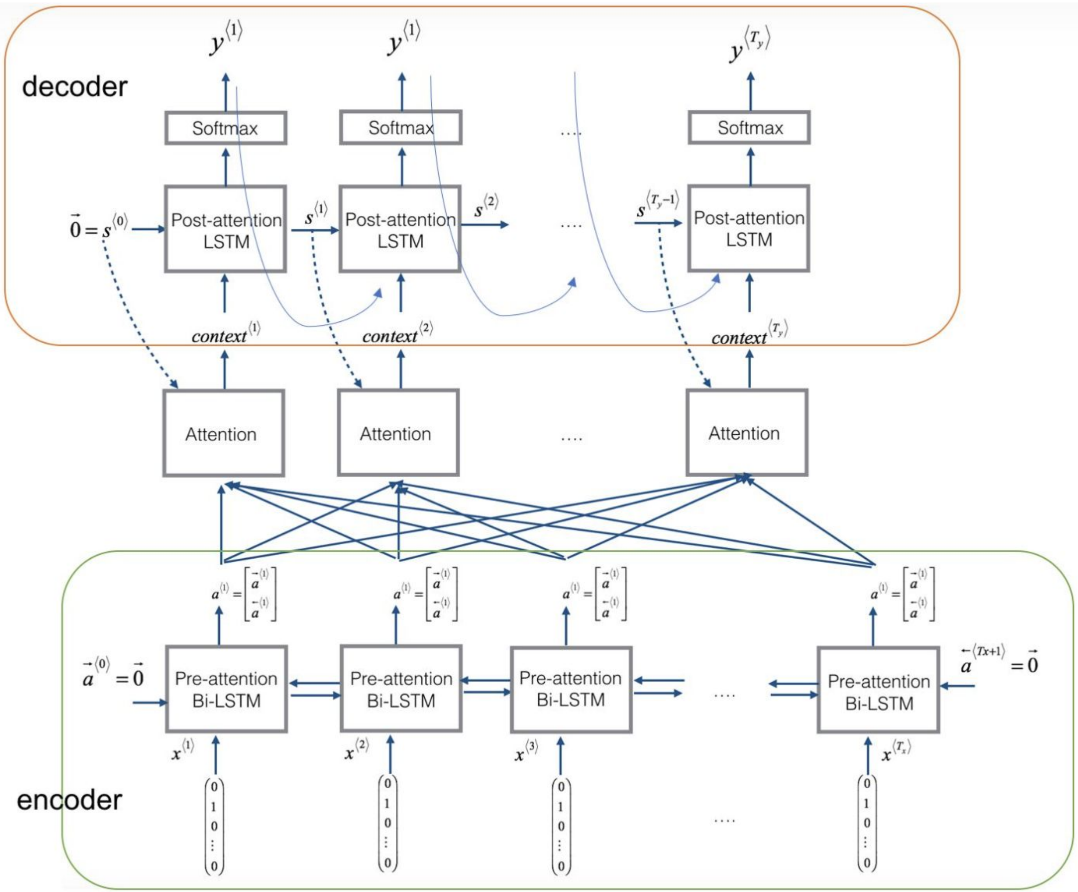
Long Short Term Memory (LSTM)

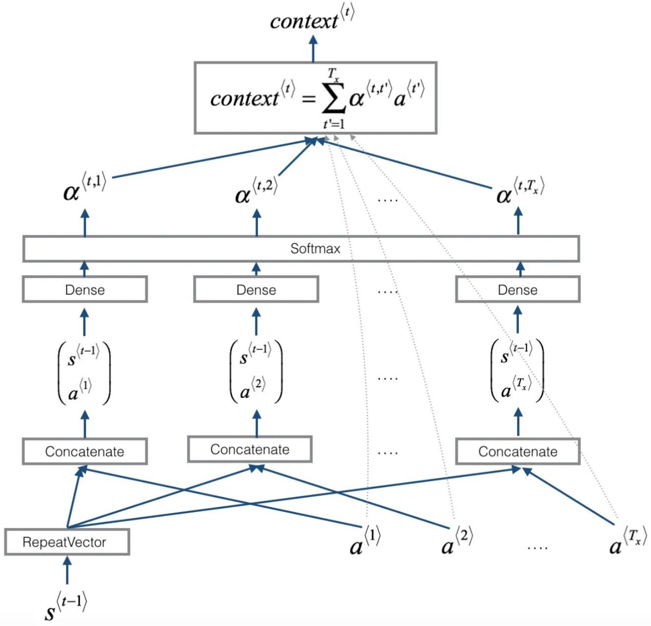


Type of gate	Role	Used in
Update gate Γ_u	How much past should matter now?	GRU, LSTM
Relevance gate Γ_r	Drop previous information?	GRU, LSTM
Forget gate Γ_f	Erase a cell or not?	LSTM
Output gate Γ_o	How much to reveal of a cell?	LSTM

	Long Short-Term Memory (LSTM)
$\tilde{c}^{<t>}$	$\tanh(W_c[\Gamma_r \star a^{<t-1>}, x^{<t>}] + b_c)$
$c^{<t>}$	$\Gamma_u \star \tilde{c}^{<t>} + \Gamma_f \star c^{<t-1>}$
$a^{<t>}$	$\Gamma_o \star c^{<t>}$

Bi-LSTM + Attention





Neural Machine Translation by Jointly Learning to Align and Translate

2014, Dzmitry Bahdanau, KyungHyun Cho, Yoshua Bengio

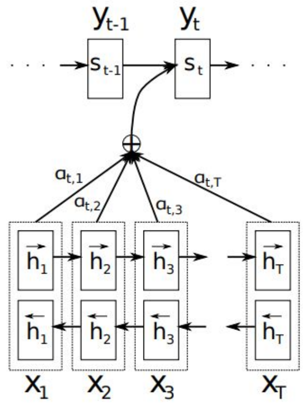
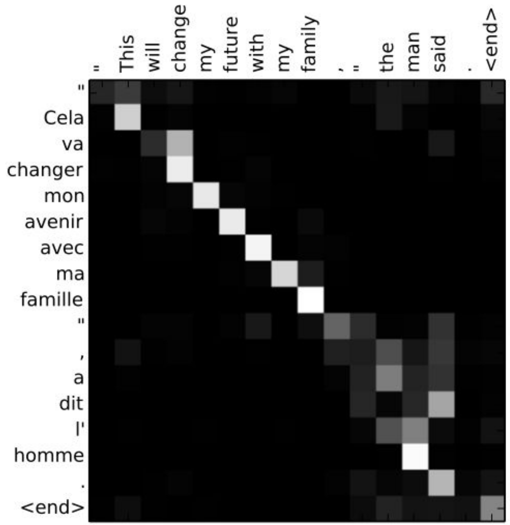


Figure 1: The graphical illustration of the proposed model trying to generate the t-th target word y_t given a source sentence $(x_1, x_2, \ldots, x_T)$.



(d)

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attention 1/3

The probability α_{ij}, or its associated energy e_{ij}, reflects the importance of the annotation h_j with respect to the previous hidden state s_{i-1} in deciding the next state s_i and generating y_i. Intuitively, this implements a mechanism of attention in the decoder. The decoder decides parts of the source sentence to pay attention to. By letting the decoder have an attention mechanism, we relieve the encoder from the burden of having to encode all information in the source sentence into a fixed-length vector. With this new approach the information can be spread throughout the sequence of annotations, which can be selectively retrieved by the decoder accordingly.

Transformers

Agenda

1. Transformer overview: 5 mins
2. Attention : 20 mins
    a. Self Attention
    b. Multi headed attention
    c. Masked Attention
    d. Cross Attention
3. Encoder architecture: 10 mins
    a. Feed forward layer
    b. Positional Encoding
    c. Residual connection
    d. Layer normalization
4. Decoder architecture: 5 mins
    a. Linear and softmax layers

Total : 40 mins

Overview

Transformer Overview

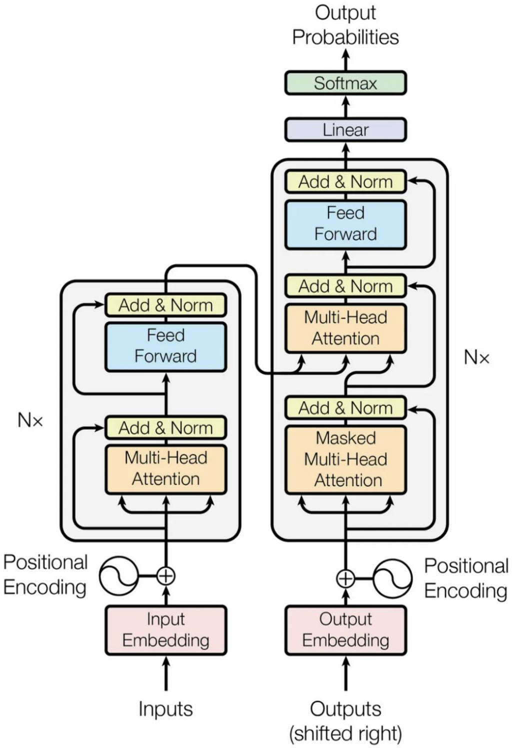

Figure 1: The Transformer - model architecture.

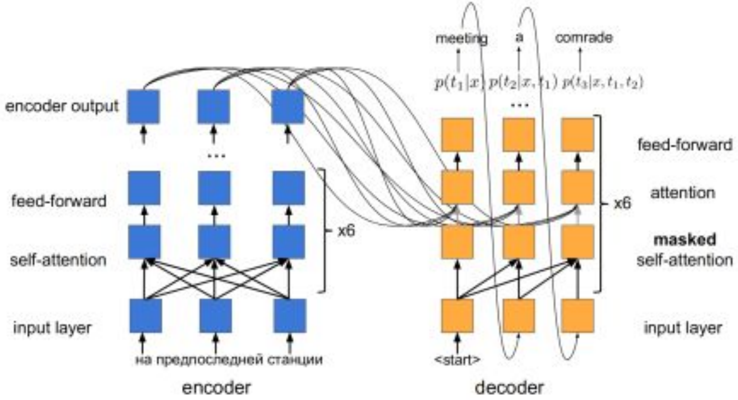

Fig. 12. Unfolded network architecture of the transformer [13]

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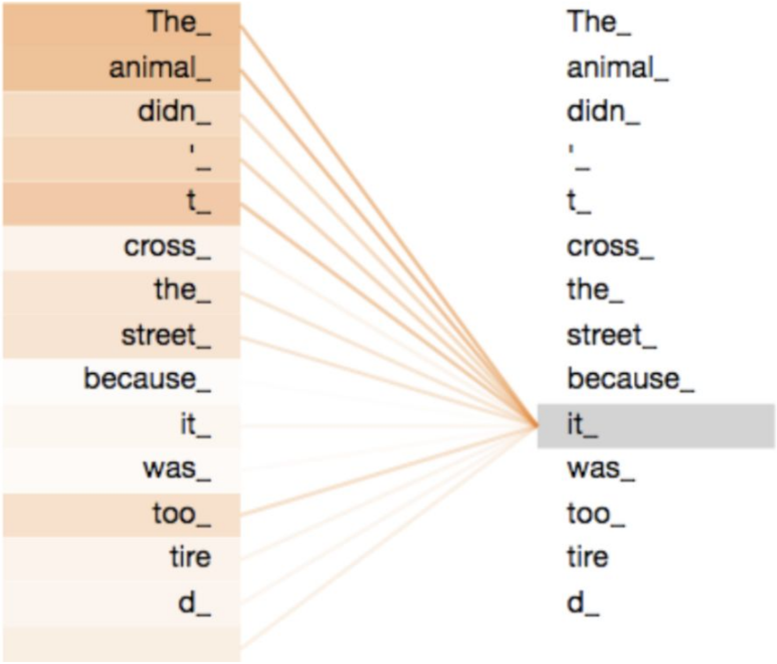
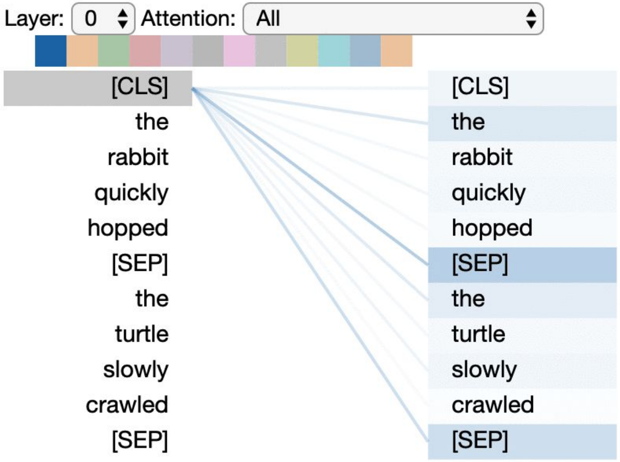
Attention

Self Attention

Learn meaning of a token through it's context





Attention Visualizations

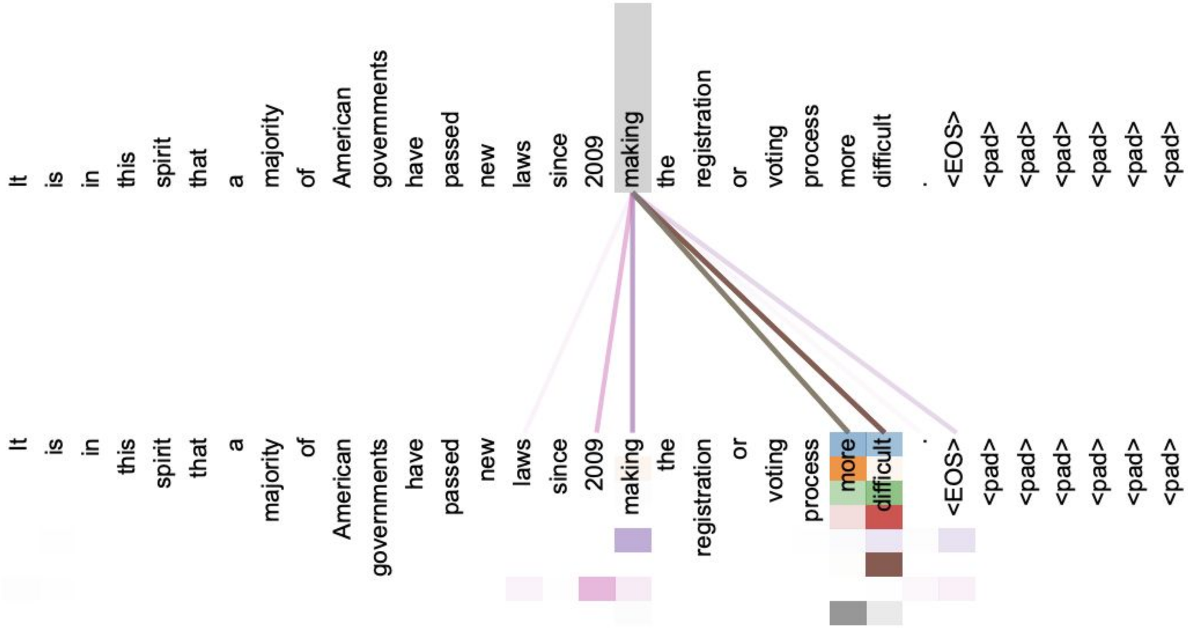
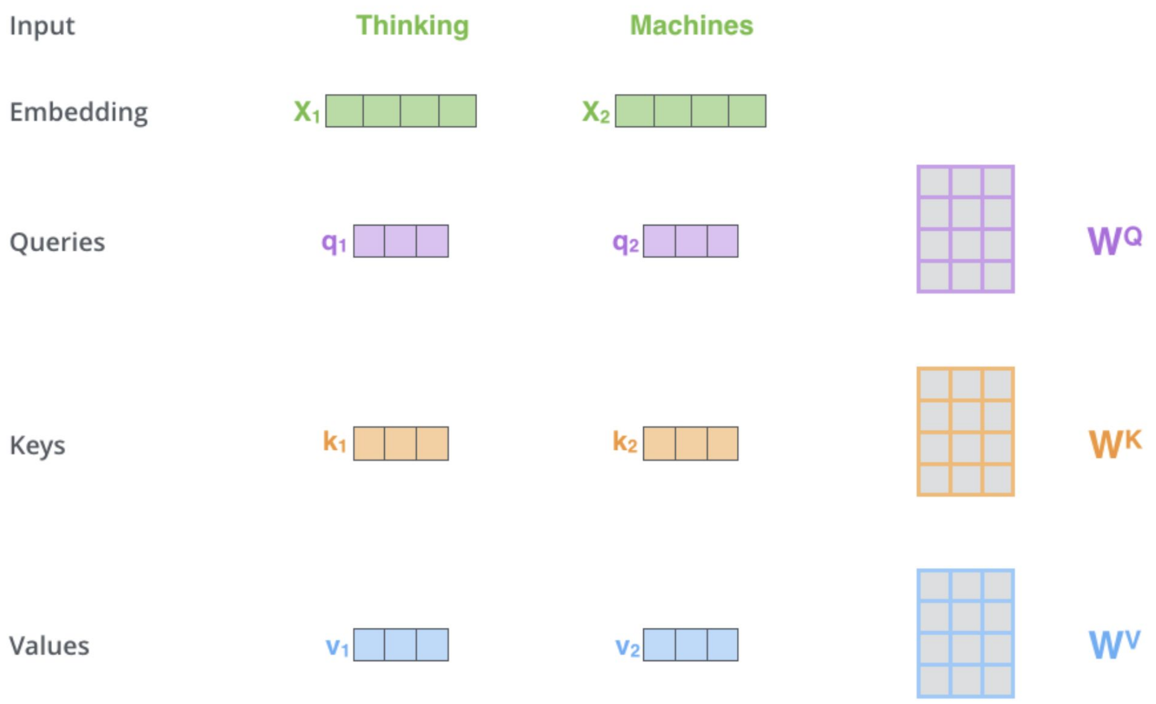


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb 'making', completing the phrase 'making...more difficult'. Attentions here shown only for the word 'making'. Different colors represent different heads. Best viewed in color.

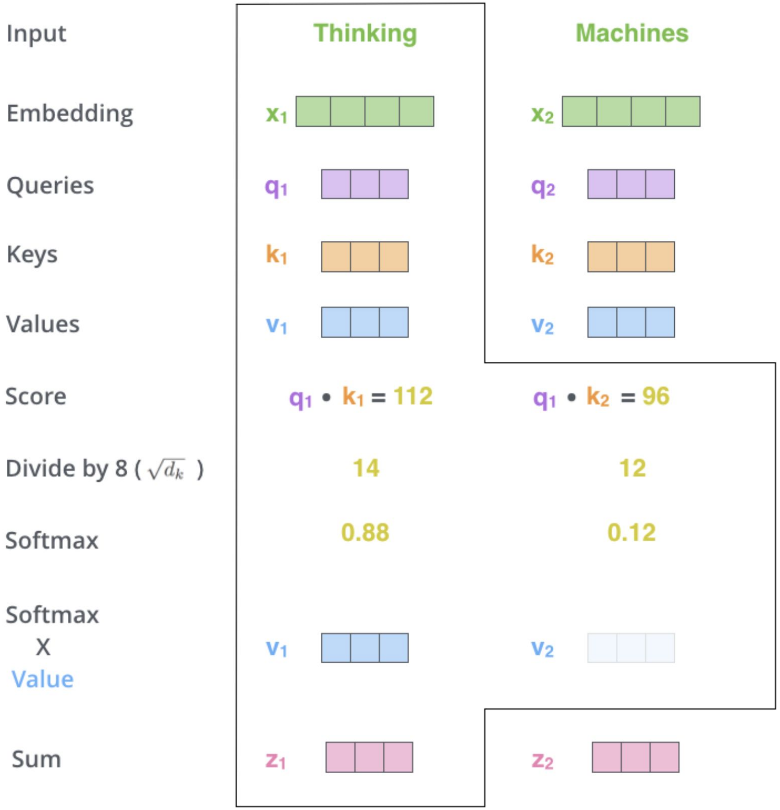
Self Attention

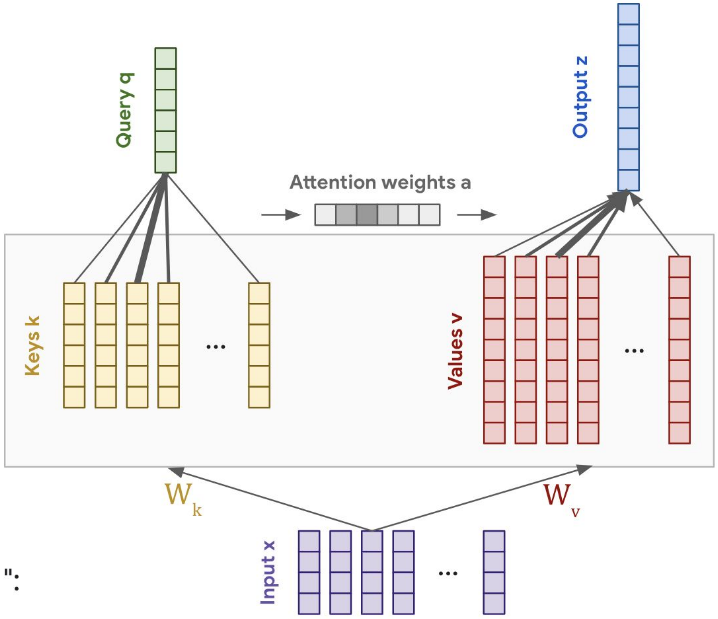
Queries, keys and values



$$Q = W_q \cdot X$$
$$K = W_k \cdot X$$
$$V = W_v \cdot X$$

Self Attention





Self Attention

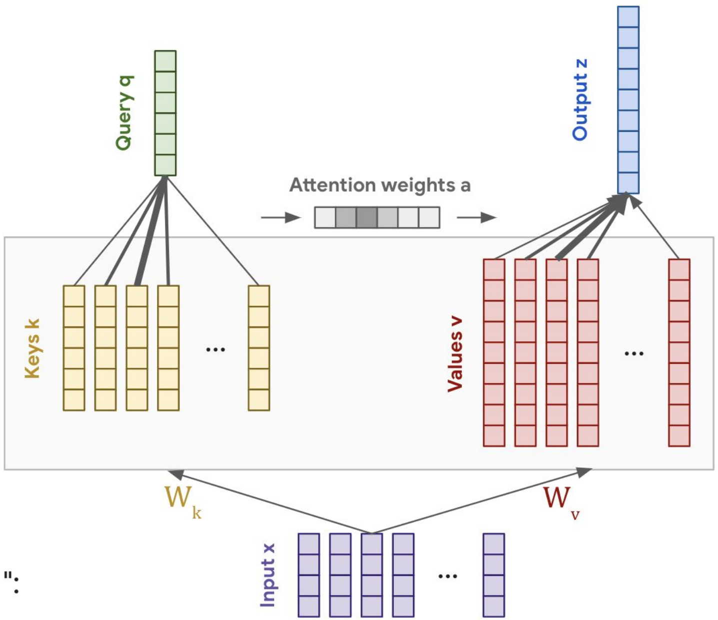
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

1. Attention weights $a_{1:N}$ are query-key similarities:

   $\hat{a}_i = q \cdot k_i$

2. Output z is attention-weighted average of values $v_{1:N}$

   $z = \sum_i \hat{a}_i v_i = \hat{a} \cdot v$



Multi-Headed Attention

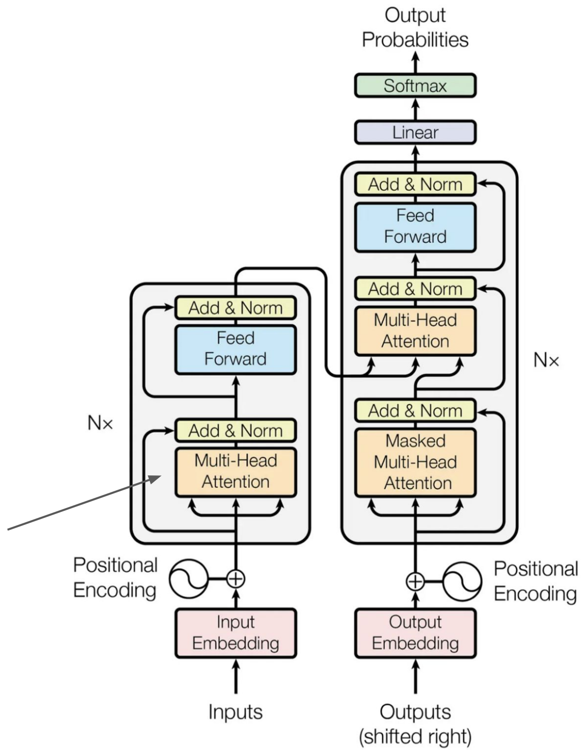
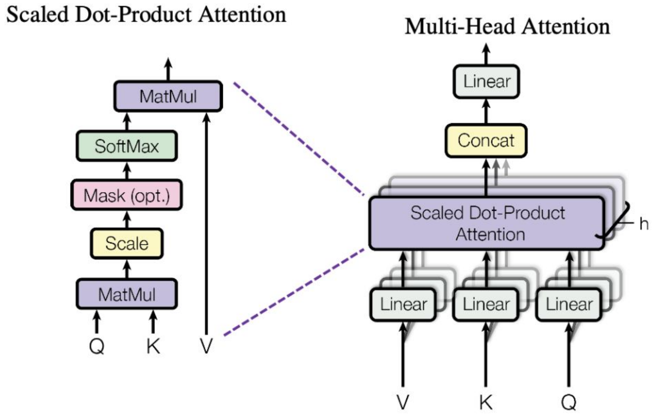
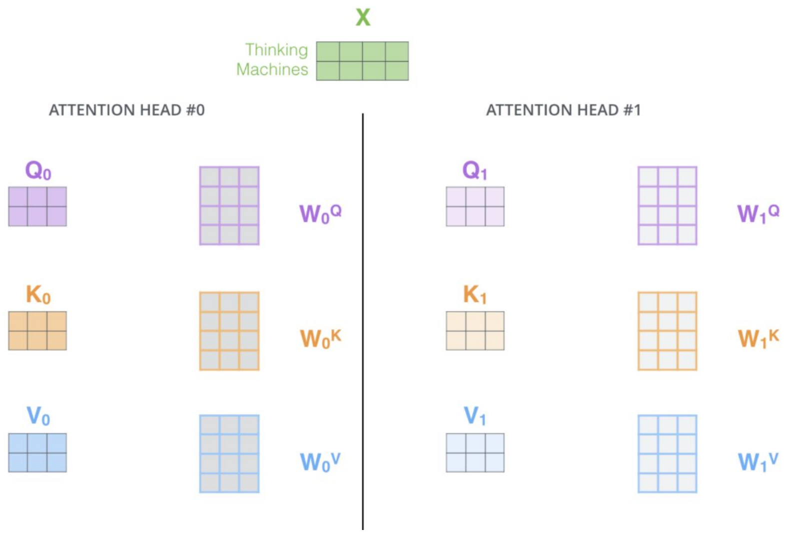


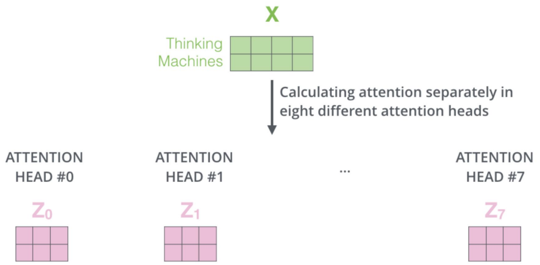
Figure 1: The Transformer - model architecture.

Multi-Headed Attention

$$\mathrm{MultiHead}(Q,K,V) = \mathrm{Concat}(\mathrm{head_1}, \ldots, \mathrm{head_h})W^O$$
$$\text{where } \mathrm{head_i} = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$







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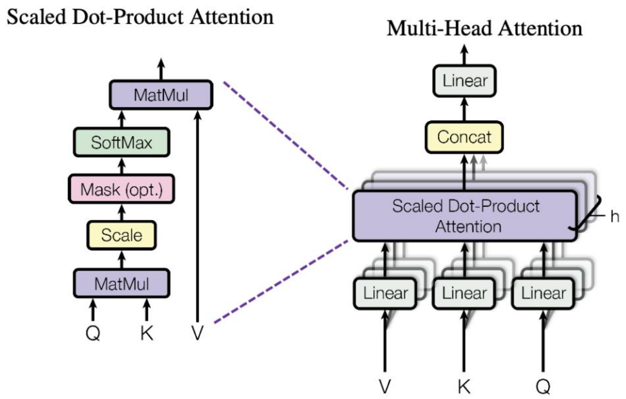
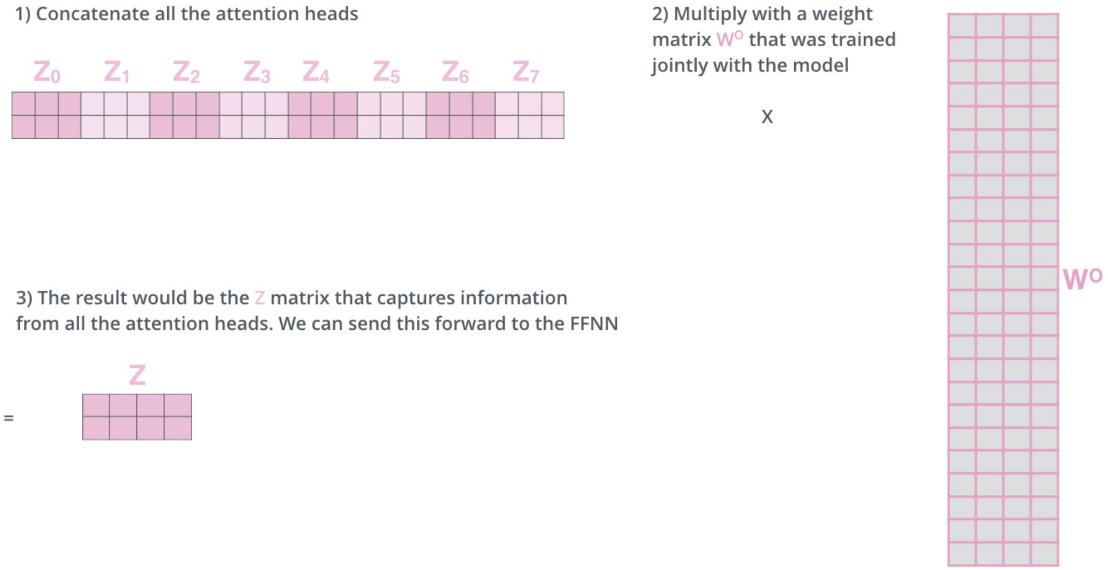
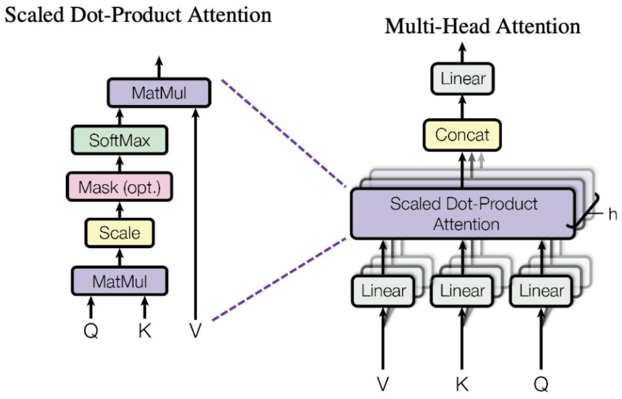
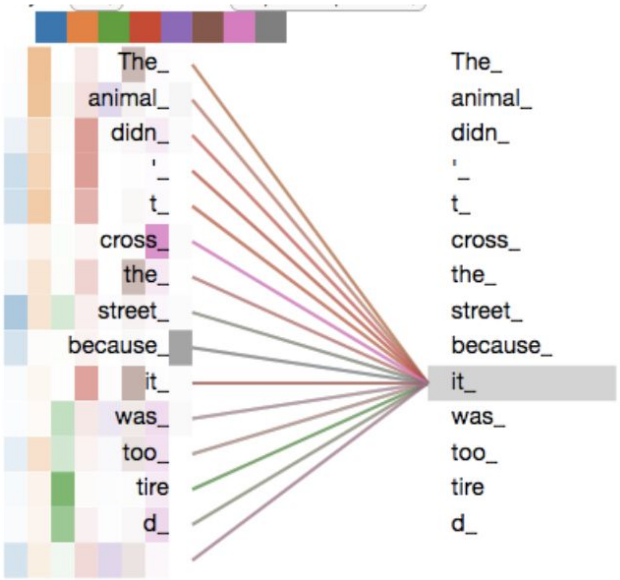


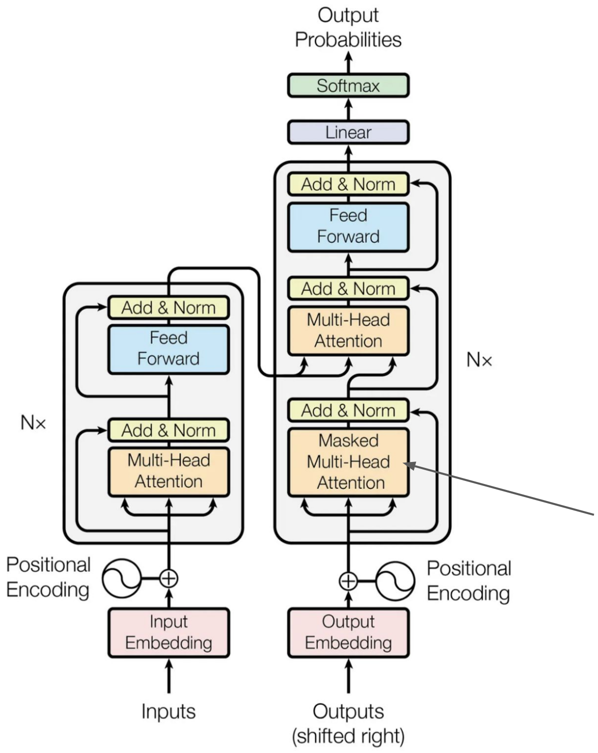


Multi-Headed Attention





Masked Multi-Headed Attention



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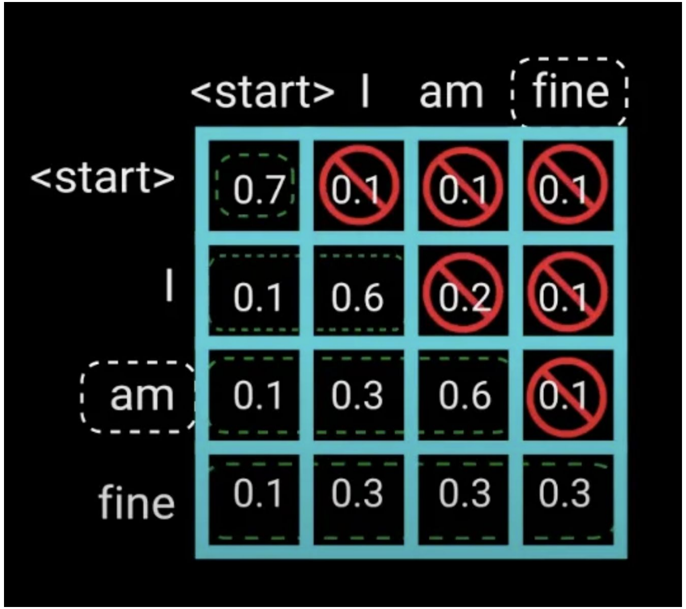
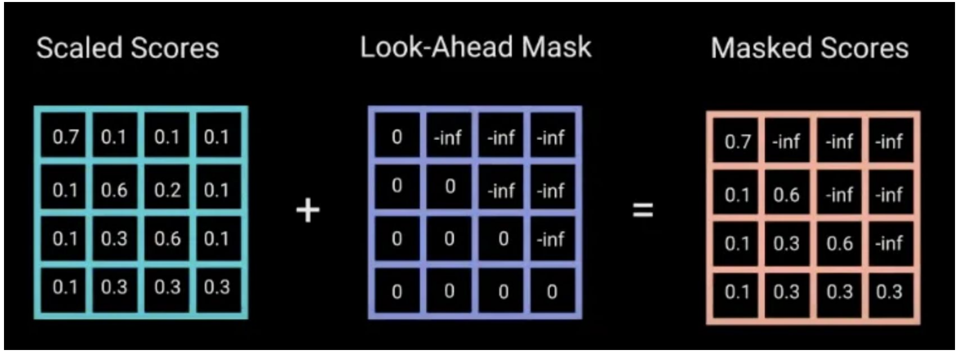
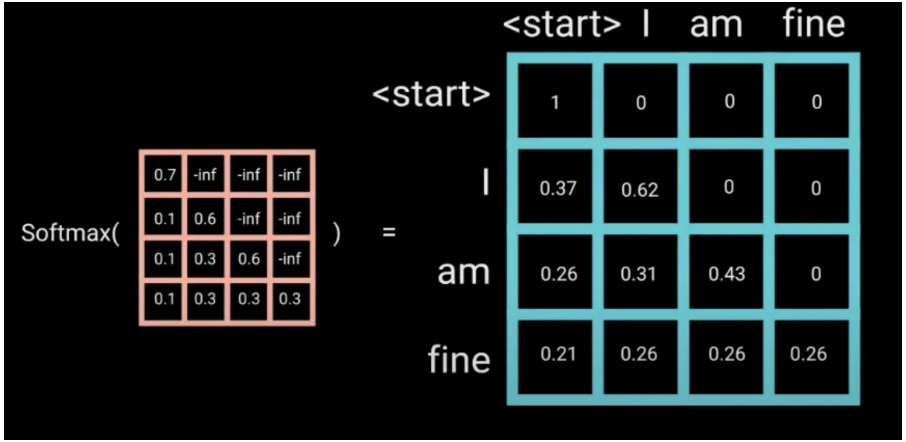
Masked Multi-Headed Attention

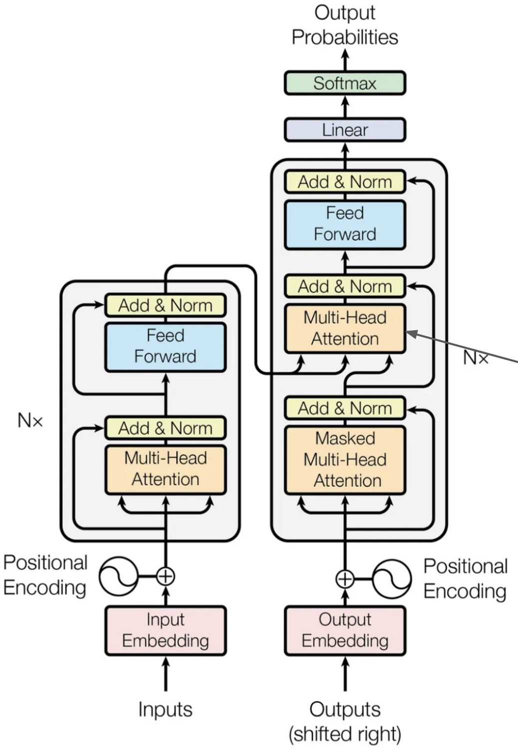
Mask future tokens while decoding







Cross Attention



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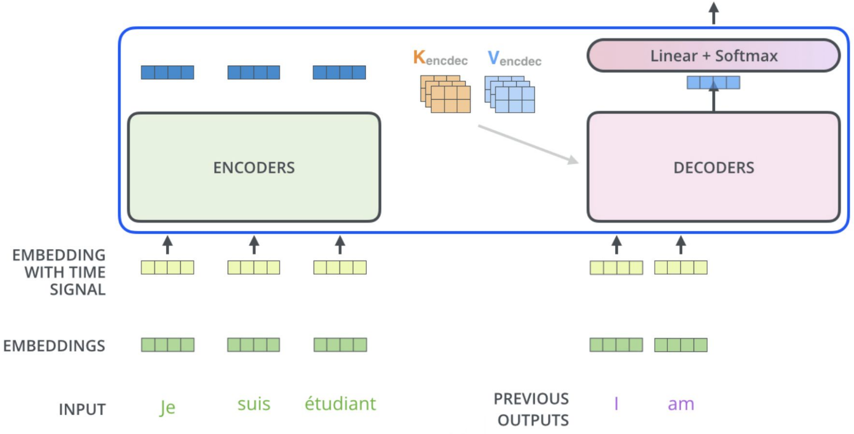

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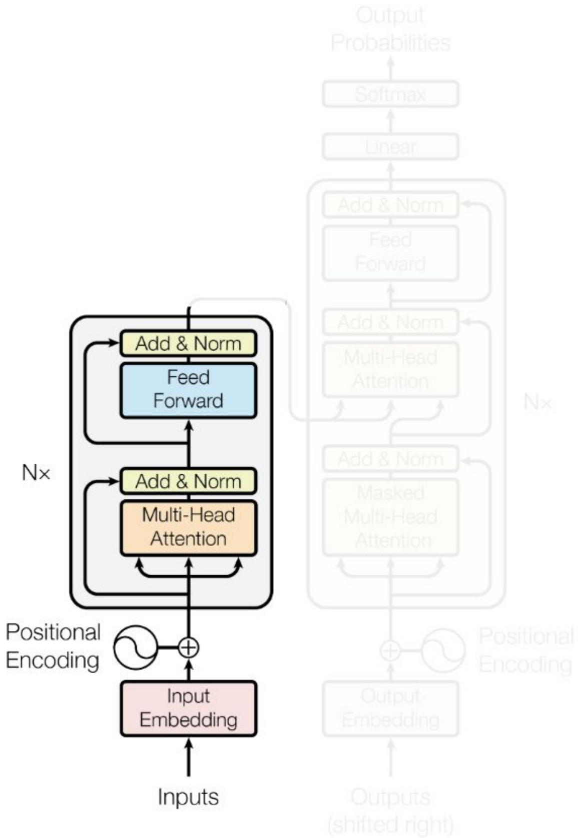
Total : 40 mins

Encoder Architecture

Encoder Architecture

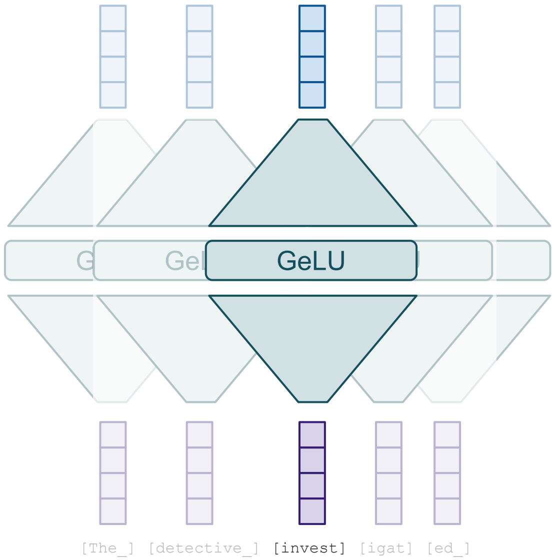


- Feed forward layer
- Positional Encoding
- Residual connection
- Layer normalization

Feed Forward Layer

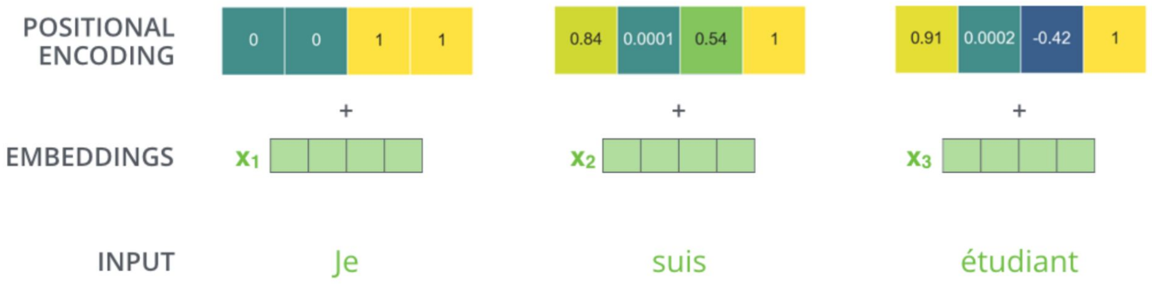
$$z_i = W_2\,\mathrm{ReLU}(W_1 x + b_1) + b_2$$

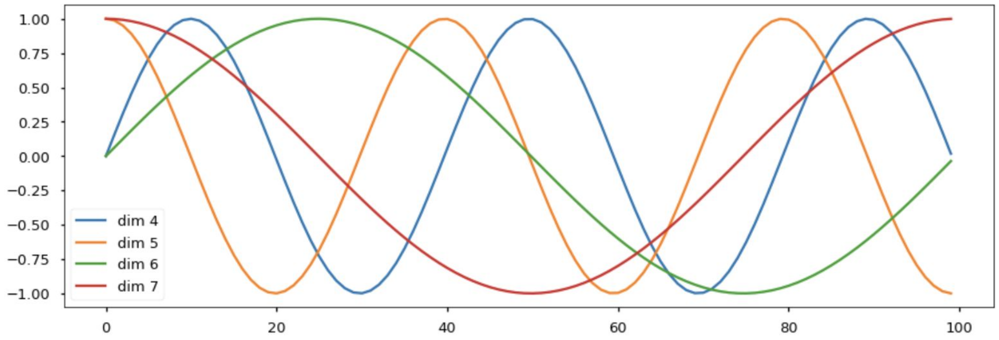
- It contains the bulk of the parameters.
- Two-thirds of a transformer model's parameters
- There's some weak evidence this is where "world knowledge" is stored, too
- Input: 512, hidden layer: 2048, output: 512



GeLU: Gaussian Error Linear Unit

Positional Encoding

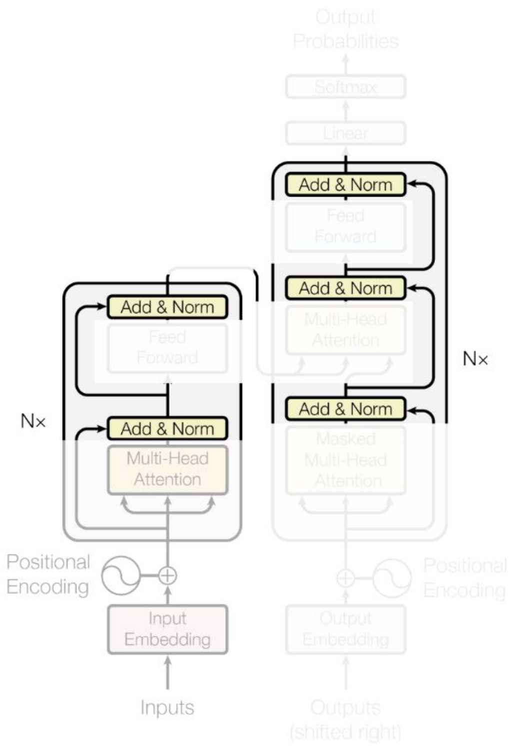




$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

Encoder Architecture



Residual connections

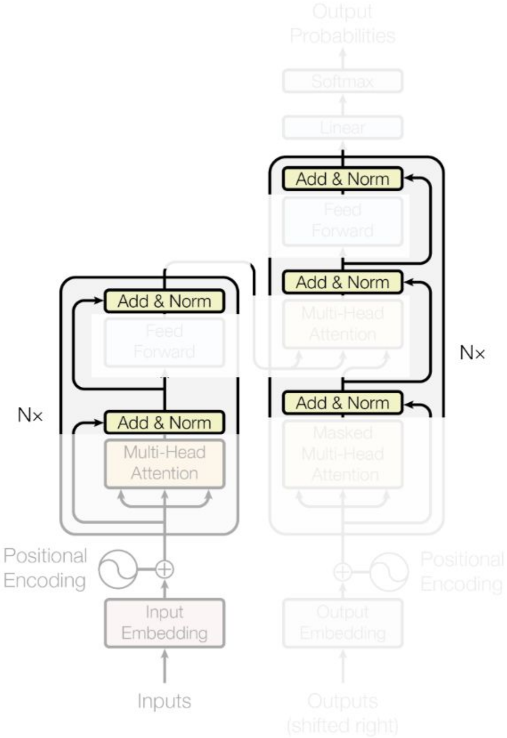
Each module's output has the exact same shape as its input.

Following ResNets, the module computes a "residual" instead of a new value:

$$z_i = \mathrm{Module}(x_i) + x_i$$

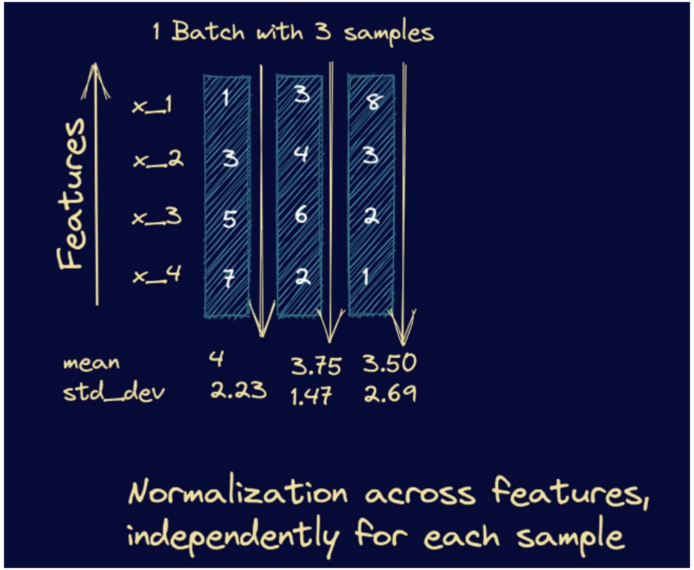
This was shown to dramatically improve trainability.

Encoder Architecture



Layer Normalization

Normalization also dramatically improves trainability



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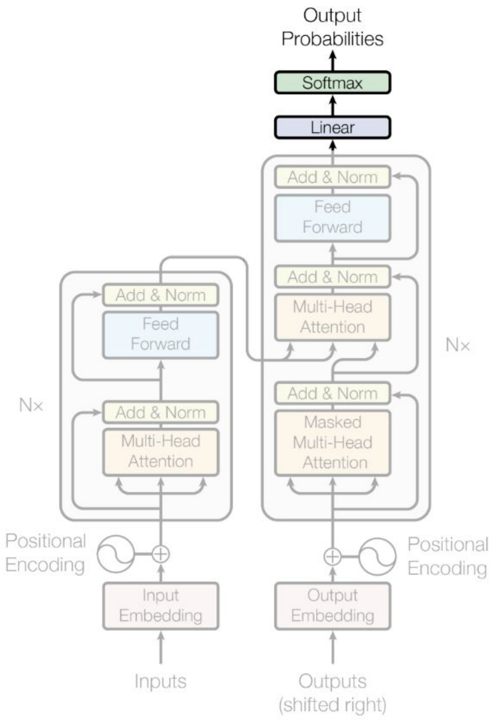
Decoder Architecture

Decoder Architecture



Encoding - Decoding

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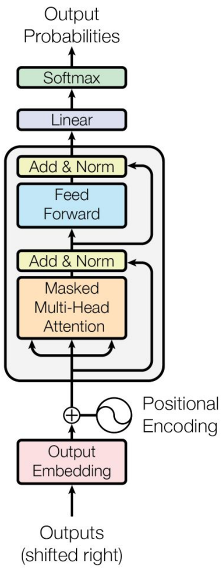
Generalizable Architecture

Decoder-only GPT

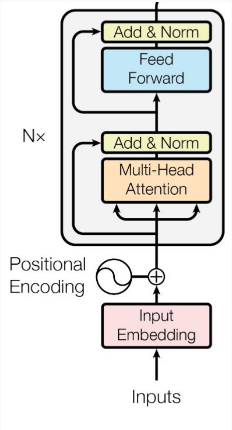
[sat_]



[START] [The_] [cat_]

Encoder-only BERT

[*] [*] **[sat_]** [*] **[the_]** [*]



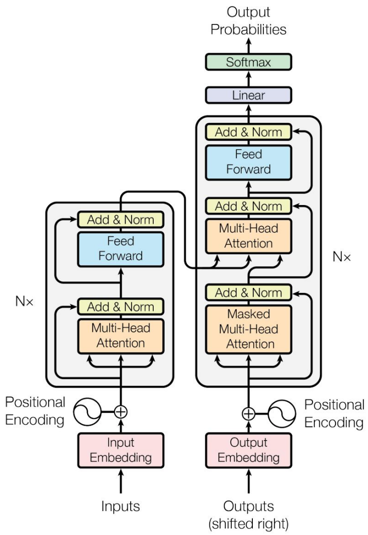
[The_] [cat_] **[MASK]** [on_] **[MASK]** [mat_]

Enc-Dec T5

Das ist gut.

A storm in Attala caused 6 victims.

This is not toxic.



Translate EN-DE: This is good.

Summarize: state authorities dispatched…

Is this toxic: You look beautiful today!

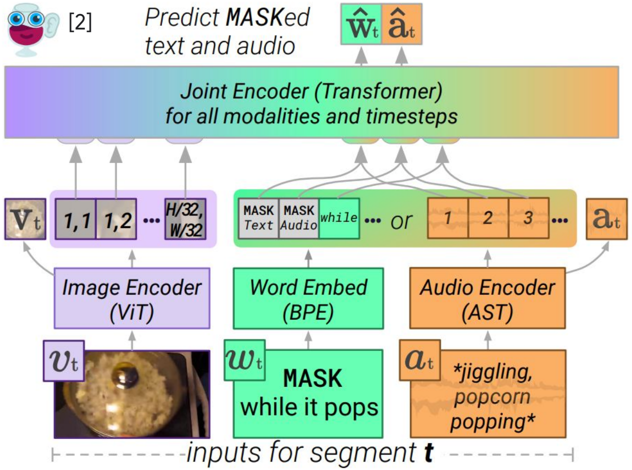
Transformer image source: "Attention Is All You Need" paper

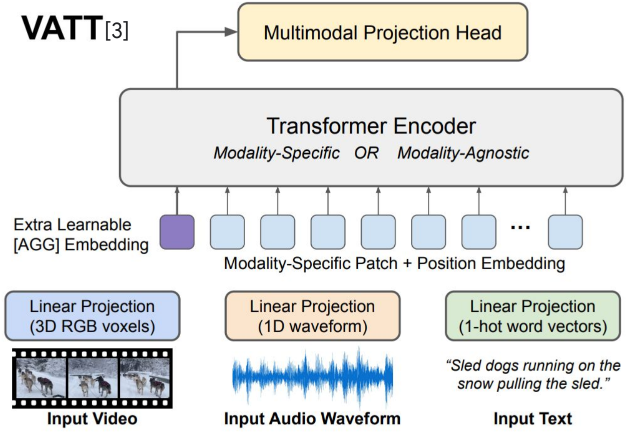
Anything you can tokenize, you can feed to Transformer

ca 2021 and onwards

Tokenize different modalities each in their own way (some kind of "patching"), and send them all jointly into a Transformer...

Seems to just work...
Currently an explosion of works doing this!





[1]

Images from:
[1] LIMoE by B Mustafa, C Riquelme, J Puigcerver, R Jenatton, N Houlsby
[2] MERLOT Reserve by R Zellers, J Lu, X Lu, Y Yu, Y Zhao, M Salehi, A Kusupati, J Hessel, A Farhadi, Y Choi
[3] VATT by H Akbari, L Yuan, R Qian, W-H Chuang, S-F Chang, Y Cui, B Gong

Thanks for your...

Attention